

WASP-104b

R-2245

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_221224 · created 2026-08-15T23:25:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459937.9965 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.044
Transit depth (Rp/Rs)^2	2.17 +/- 1.29 [%]
Orbital Inclination (inc)	82.34 +/- 2.87
Airmass coefficient 1 (a1)	2.81 +/- 1.46
Airmass coefficient 2 (a2)	-0.95 +/- 1.62
Scatter in the residuals of the lightcurve fit is	51.44 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	3.0
Transit Duration (day)	0.043 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.044 vs 0.1214 (deviation 0.58σ)
Duration fit vs published	0.043 d vs 0.0742 d (deviation 1.25σ)
Mid-time O–C	11.9 min
Depth SNR	3.3
Residual scatter	51.44 %

Light curve

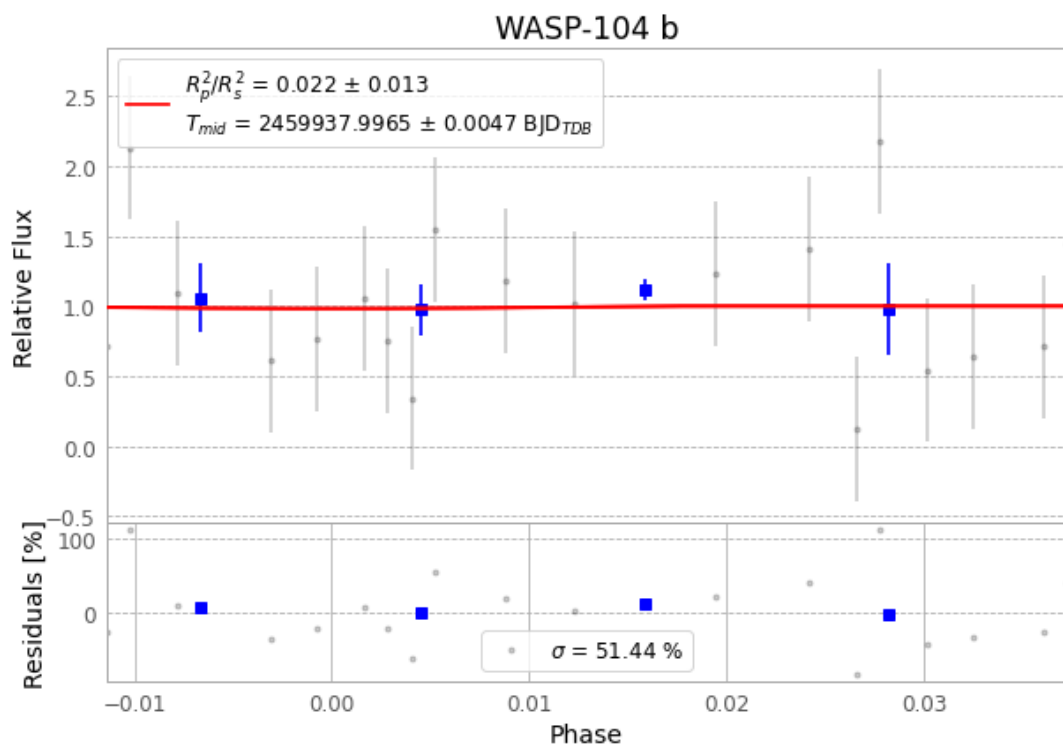


Figure 1 — FinalLightCurve_WASP-104 b_2022-12-24.png (SHA-256 4cbdbece3decfd49...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 97], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ed53851ff9c10e3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

42 FITS frames (28 MB), WASP-104221224112115.FITS → WASP-104221224132415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-221224.log (fc8f39c91c78...), FinalLightCurve_WASP-104 b_2022-12-24.png (4cbdbece3dec...), FinalLightCurve_WASP-104 b_2022-12-24.csv (aba1fbe6fecf...)

Compute resources

Wall time 82.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:25Z.